

Lab 07

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1. Baixe o arquivo `uwmadison.sqlite3`. Conecte-se a ele usando o pacote `RSQLite`, armazenando a conexão em uma variável `conn`.

```
library(DBI)
library(RSQLite)

conn <- dbConnect(RSQLite::SQLite(), "./database.sqlite3")
```

```
dbListTables(conn)
```

```
[1] "course_offerings"  "courses"           "grade_distributions"
[4] "instructors"       "rooms"              "schedules"
[7] "sections"          "subject_memberships" "subjects"
[10] "teachings"
```

2. Quem são e quantos são os professores que lecionaram disciplinas cujo tópico era estatística (`subjects.abbreviation='STAT'`);

```
SELECT DISTINCT i.name AS lista_professores
FROM instructors i
JOIN teachings t ON i.id = t.instructor_id
JOIN sections s ON t.section_uuid = s.uuid
JOIN course_offerings co ON s.course_offering_uuid = co.uuid
JOIN subject_memberships sm ON sm.course_offering_uuid = co.uuid
JOIN subjects sub ON sm.subject_code = sub.code
WHERE sub.abbreviation = 'STAT'
ORDER BY i.name;
```

Table 1: Displaying records 1 - 10

lista_professores
ABIGAIL BENZINE
ADIN-CRISTIAN ANDREI
AKICHIKA OZEKI
ALAN HUANG
ALBERTO DEL PIA
ALBRECHT KLEMM
ALEXANDER COVINGTON
ALEXANDER FISH
ALEXANDER KISELEV
ALYSSA DIGILIO

3. O GPA americano é definido numa escala de 0 a 4, em que A = 4, AB = 3.5, B = 3, BC = 2.5, C = 2, D = 1 e F = 0. Determinando a nota média de cada oferecimento pela ponderação da quantidade de alunos em cada extrato com os valores numéricos de cada conceito, indique (no que se referente a disciplinas no assunto de estatística):

```
SELECT
  co.uuid AS course_offering_uuid,
  co.name AS nome_curso,
  (SUM(gd.a_count * 4.0 + gd.ab_count * 3.5 + gd.b_count * 3.0 + gd.bc_count * 2.5 + gd.c_count * 2.0 + gd.d_count * 1.0 + gd.f_count * 0.0) /
   NULLIF(SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count + gd.d_count + gd.f_count), 0)) AS media_gpa
FROM course_offerings co
JOIN subject_memberships sm ON co.uuid = sm.course_offering_uuid
JOIN subjects sub ON sm.subject_code = sub.code
JOIN grade_distributions gd ON co.uuid = gd.course_offering_uuid
WHERE sub.abbreviation = 'STAT'
GROUP BY co.uuid, co.name;
```

Table 2: Displaying records 1 - 10

course_offering_uuid	nome_curso	media_gpa
00fec14b-65ce-32bb-87d7-a9c943b7fa2e	Intro to Mathematical Stat	2.826087
011d9daf-c271-3c62-90f4-a3f752b8547c	Senior Honors Thesis	NA
01ab0af6-7627-36aa-a2c4-3db1d80554c4	Applied Regression Analysis	3.078947
01d286a6-691b-383a-9822-0172411bd420	Intro-Stochastic Processes	3.116667
0242fc43-7318-3afb-be81-003021df769d	Intro to Clinical Trials I	3.615385

course_offering_uuid	nome_curso	media_gpa
02e9d068-6914-3413-899b-68e58277d9dc	Statistical Consulting	NA
03200862-7601-3098-8e5c-38f8cd835a1c	Intro Applied Stats for Engrs	3.160919
035f6a64-d676-36dd-b4df-b4fb6666dedf	Directed Study	NA
03a155a1-32c6-3411-b388-9e59cea81c2c	Intro Stats for Engineers	2.871875
03a9c506-e84e-32be-809b-2a627d59039b	Statistical Consulting	4.000000

4. Quem é o professor mais difícil? Quem é o professor mais fácil? Qual é a disciplina mais difícil? Qual é a disciplina mais fácil?

Professor Mais Difícil (Menor GPA):

```
SELECT
    i.name AS lista_professores,
    (SUM(gd.a_count * 4.0 + gd.ab_count * 3.5 + gd.b_count * 3.0 + gd.bc_count * 2.5 + gd.c_count * 2.0 + gd.d_count * 1.0) /
    NULLIF(SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count + gd.d_count), 0)) AS media_gpa
FROM instructors i
JOIN teachings t ON i.id = t.instructor_id
JOIN sections s ON t.section_uuid = s.uuid
JOIN course_offerings co ON s.course_offering_uuid = co.uuid
JOIN subject_memberships sm ON sm.course_offering_uuid = co.uuid
JOIN subjects sub ON sm.subject_code = sub.code
JOIN grade_distributions gd ON co.uuid = gd.course_offering_uuid
WHERE sub.abbreviation = 'STAT'
GROUP BY i.id, i.name
HAVING SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count + gd.d_count) > 0
ORDER BY media_gpa
LIMIT 1
```

Table 3: 1 records

lista_professores	media_gpa
JAMES D KUELBS	2.607692

Professor mais fácil (Maior GPA):

```
SELECT
    i.name AS lista_professores,
    (SUM(gd.a_count * 4.0 + gd.ab_count * 3.5 + gd.b_count * 3.0 + gd.bc_count * 2.5 + gd.c_count * 2.0 + gd.d_count * 1.0) /
    NULLIF(SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count + gd.d_count), 0)) AS media_gpa
FROM instructors i
JOIN teachings t ON i.id = t.instructor_id
JOIN sections s ON t.section_uuid = s.uuid
JOIN course_offerings co ON s.course_offering_uuid = co.uuid
JOIN subject_memberships sm ON sm.course_offering_uuid = co.uuid
JOIN subjects sub ON sm.subject_code = sub.code
JOIN grade_distributions gd ON co.uuid = gd.course_offering_uuid
WHERE sub.abbreviation = 'STAT'
GROUP BY i.id, i.name
HAVING SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count + gd.d_count) > 0
ORDER BY media_gpa
LIMIT 1
```

```

    NULLIF(SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count + gd.d_count
FROM instructors i
JOIN teachings t ON i.id = t.instructor_id
JOIN sections s ON t.section_uuid = s.uuid
JOIN course_offerings co ON s.course_offering_uuid = co.uuid
JOIN subject_memberships sm ON sm.course_offering_uuid = co.uuid
JOIN subjects sub ON sm.subject_code = sub.code
JOIN grade_distributions gd ON co.uuid = gd.course_offering_uuid
WHERE sub.abbreviation = 'STAT'
GROUP BY i.id, i.name
HAVING SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count + gd.d_count + g
ORDER BY media_gpa DESC
LIMIT 1

```

Table 4: 1 records

lista_professores	media_gpa
SRIKANTHMADHAVAN ARAVAMUTHAN	4

Disciplina Mais Difícil (Menor GPA):

```

SELECT
    co.name AS nome_curso,
    (SUM(gd.a_count * 4.0 + gd.ab_count * 3.5 + gd.b_count * 3.0 + gd.bc_count * 2.5 + gd.c_
    NULLIF(SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count + gd.d_count
FROM course_offerings co
JOIN subject_memberships sm ON co.uuid = sm.course_offering_uuid
JOIN subjects sub ON sm.subject_code = sub.code
JOIN grade_distributions gd ON co.uuid = gd.course_offering_uuid
WHERE sub.abbreviation = 'STAT'
GROUP BY co.uuid, co.name
HAVING SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count + gd.d_count + g
ORDER BY media_gpa
LIMIT 1

```

Table 5: 1 records

nome_curso	media_gpa
Intro-Stochastic Processes	2.441177

Disciplina Mais Fácil (Maior GPA):

```
SELECT
  co.name AS nome_curso,
  (SUM(gd.a_count * 4.0 + gd.ab_count * 3.5 + gd.b_count * 3.0 + gd.bc_count * 2.5 + gd.c_count * 2.0 + gd.d_count * 1.0)
   NULLIF(SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count + gd.d_count))) AS media_gpa
FROM course_offerings co
JOIN subject_memberships sm ON co.uuid = sm.course_offering_uuid
JOIN subjects sub ON sm.subject_code = sub.code
JOIN grade_distributions gd ON co.uuid = gd.course_offering_uuid
WHERE sub.abbreviation = 'STAT'
GROUP BY co.uuid, co.name
HAVING SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count + gd.d_count) > 0
ORDER BY media_gpa DESC
LIMIT 1
```

Table 6: 1 records

nome_curso	media_gpa
Statistical Consulting	4

5. Desconecte do banco de dados.

```
dbDisconnect(conn)
```

```
Sys.setenv(TZ = "America/Sao_Paulo")
Sys.time()
```

```
[1] "2026-09-23 22:11:34 -03"
```